S475/1

SUBSIDIARY

MATHEMATICS

Paper 1

JULY/AUGUST 2026

$2\frac{1}{2}$ hours

JINJA JOINT EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

MOCK EXAMINATIONS –JULY / AUGUST, 2026

SUBSIDIARY MATHEMATICS

Paper 1

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

This paper consists of two sections; **A** *and* **B.**

Section **A** *is* ***compulsory.***

Section **B** *consists of* **two** *parts;* **one** *and* **two.** *Answer only* **two** *questions from this section, choosing* ***only one*** question from each part.

Any additional question(s) answered will **not** *be marked.*

All *necessary working* **must** *be shown clearly.*

Begin each answer on a fresh sheet of paper.

Graph paper is provided.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

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Section: A (compulsory)

Item One

Two drones were deployed by government to drop food packages to specific places in Bududa district specifically to those people who had been affected by landslides.

These drones had one control room which had coordinates C(2,5) on a google map.

Drones A and B were sent to places with coordinates on a Google map as A(4, 8) and B(15, 11) kilometers respectively.

For the deliveries to be made successfully, the angle between the drones relative to the control room must be at least 30^0. The drone controller would like to know whether the deliveries were successful or not.

In this control room, the drone controller monitors the flights of the drones using a signal transmitter which has two antennae attached to it at the same point. The signal strength depends on the orientation angle θ of the transmitter's antennae relative to the tower. To have a high signal strength, the orientation angle between the two antennae should be at most 60^0; $(0^0 \leq \theta \leq 90^0)$.

The drone controller modelled the signal strength(S) as $-2sin^2\theta - 3sin\theta + 3.5 = 0$. The drone controller would like to know whether this model can produce high signal strength or not.

Task:

With mathematical justifications, help the drone controller to get the required information.

Item Two

GML construction company has been contracted by Joyo town council to renovate the curved road that connects the town council headquarters to the district headquarters. The road to be renovated has many corners which must be reduced.

The road is modelled by the equation $y = x^4 - 3x^2 + 5x - 2$ and the company manager has decided to reduce all the corners near those points with slopes more than 9 since many vehicles fall there. These corners are near points A (0,1), B (2,6) and C (-1,5). The company manager would like to know the corner(s) to be reduced.

This company has also been contracted by the town council mayor to construct for them a parking lot for motorcycles which is curved with a mathematical relationship $y = x^2 - 11x + 10$ and enclosed by a straight road given by $y = 0$. In their plan, each motorcycle will be occupying an area of $2.025m^2$.One day, the mayor received a call informing him that he is going to visited by 70 people each with a motorcycle. The mayor would like to know whether the parking lot will be enough or not.

After the mayor receiving the phone call, he started informing the community members about the visitors coming. The rate at which the number of people getting information was increasing is proportional to the number of people who had heard the information at any time t-hours and the proportionality constant was 1.2. Initially 2 people had heard the information and they agreed to start the meeting when the number of people with the information reaches 20. The mayor would like to know when the meeting will start.

Task;

(a) Advise GML construction company manager on the corners to be reduced.

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(b) By illustrating the required parking lot and with mathematical justifications help the mayor with the details he wants.

SECTION: B

Part one

Attempt only *one* item from this part

Item Three

In Kyomya west Budondo, a group of people started a savings scheme.

They set up a policy that their total monthly savings must be increasing by a certain amount.

In January 2026, they saved a total of UGX. 500,000, in February, they saved UGX. 600,000, in March 2026, they saved UGX. 700,000. They agreed to buy a plot of land worth UGX. 10,000,000 using their cumulative savings but they don't know when it will be possible.

To run the scheme successfully, the members agreed to form an executive committee of four people from among themselves with at least a member of each gender on the committee but they don't know the possible number of ways of selecting this committee.

They also decided that a new member willing to join the scheme, must contribute $UGX.20000(2\log x^2 - \log x^4 + 6\log 10)$ where x is the savings growth factor.

Mrs. Munaaba has UGX. 97,000 and she would like to join the scheme but she is not sure whether she will be allowed or not.

Task;

With mathematical justifications, help the members and Mrs. Munaaba with a detailed analysis of the required information.

Item Four

Mukasa received a government youth livelihood grant and invested the funds into an intensive poultry-keeping project.

In the first month, he bought 32 improved local chicken, in the second month he bought more 64 birds and in the third month, he bought more 128 birds. Mukasa had constructed a standard poultry shelter that safely accommodates a total of 4064 birds. He would like to know when the total number of birds will hit the shelter's capacity and he has also said that if the number of months needed to hit the shelter's capacity exceeds 5, he will sell all the birds at once.

At this very poultry farm, the workers had 4 layers,2 broilers which were distinct and 5 other identical birds to put in 10 available cages. The workers were bothered about the numbers of ways in which they can arrange all the birds so that the two broilers are in the same cage.

The daily feeds needed at the farm depends on seasonal grain prices. The farm analyst modelled the feeds required as $3x^2 + (k+2)x + 12 = 0$ where k is a constant feed quantity(kg) and x is the varying feed quantity needed on the farm. The model should have equal roots not to cause a financial loss to the farm but Mukasa does not know the possible value(s) of k that can make this possible.

Task;

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As a mathematics student help Mukasa to;

(a) Have an informed decision on when the number of birds will hit the shelter's capacity and whether to sell all the birds.

(b) Determine the possible number of ways of arranging the birds in the available cages and the possible values of the constant feed quantity

Part two

Attempt only ***one*** item from this part

Item Five

At a certain rice milling machine, rice to be sold is put in sealed bags after taking their measurements. Some of bags sealed are underweight and others have standard weights from the analysis made.

It was noted that out of every 100 bags sealed, 30 bags are under weight (weigh less than 50kg) which caused a serious concern to UNBS. One day UNBS officials visited the rice store abruptly and re-measured the different bags but before re-measuring, UNBS officials agreed with the machine owner that;

(i) the license will be withdrawn if the chances of getting at least 10 bags from the 15 bags selected being underweight is greater than 0.2.

(ii) If the variance is greater than 2, then there will be a fine of UGX 700,000.

The weights of bags sealed are also normally distributed with mean weight of 48kg and standard deviation of standard deviation of 4. The machine owner decided to buy a new weighing scale if the chance of getting a bag weighing less than 45kg is greater than 0.5 otherwise he remains with the old weighing scale.

At this milling machine, the owner sold bags of unprocessed rice, processed rice and sacks of rice husks with the intention to raise UGX. 83,000,000 to buy an acre of land to expand the project.

In that particular month, he sold 40 bags of unprocessed rice, 60 bags of processed rice and 30 sacks of rice husks to Muzani. 30 bags of unprocessed rice, 50 bags of processed rice and 27 sacks of rice husks to Joel. 25 bags of unprocessed rice, 43 bags of processed rice and 32 sacks of rice husks to Hamson. 32bags of unprocessed rice, 29 bags of processed rice and 12 sacks of rice husks James. Each bag of unprocessed rice, processed rice and sack of rice husks were sold at UGX. 180,000, UGX. 280,000 and UGX. 15,000 respectively but he does not know whether the sales made will raise the required amount to buy land.

Task;

With mathematical evidences, justify the decisions to be taken by UNBS and also help the owner to determine the required information.

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Item Six

At Doho health Centre II, a team of clinical officers had an argument on whether there is a very high relationship between the heights of patients and length of the arms or not.

They selected a sample of 10 patients and took their measurements and recorded the results in the table as shown below:

Patient code	**P01**	**P02**	**P03**	**P04**	**P05**	**P06**	**P07**	**P08**	**P09**	**P10**
Height(x) cm	132	145	128	150	135	140	132	155	142	138
Length of the arm (y) cm	32	40	28	48	33	35	30	52	41	33

Their team leader told them that the relationship is said to be very high positive if the coefficient of correlation is at least 0.80 otherwise it is not very high. The clinical officers wanted to confirm whether there is a very high positive relationship between height and length of the arm using the data gathered and they also wanted to know what could be the height for a patient whose length of the arm is 46cm.

The clinical officers also wanted to determine the standard deviation for height of the patients selected with the intention to order for a new measuring tape to be used. They agreed that if the standard deviation for height exceeds 8.0, they will order for a new measuring tape otherwise they maintain the old one.

One patient brought 7 oranges and 3 lemons in a small covered bag as a gift to the clinical officers. He told one clinical officer to pick two fruits at random one at a time without returning any fruit. This clinical officer was interested in picking one lemon and one orange but he did not know the chances he had to pick what he wanted.

The health center supervisor, records the number of patients(X) received each day with a probability distribution as shown below;

Number of patients	0	1	2	3	4
$P(X = x)$	0.10	k	1.5k	0.25	0.15

The supervisor said that if $P(X \geq 2)$ is greater than 0.80, he will add one another clinical officer on duty every day otherwise he maintains the same number of clinical officers on duty.

Task;

Using mathematics justifications, help the team of clinical officers and the supervisor with the required information.

END

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